



Amazon RDS — Complete Notes

Relational Database Service · Managed SQL databases on AWS

AWS RDS Service Database Model Managed Engines 6

A reader-friendly, all-in-one reference for Amazon RDS.

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1. What is RDS

Amazon RDS is a **managed relational database** service. AWS handles the heavy operational work — provisioning, OS and DB patching, backups, replication, failover, and scaling — so you focus on schema and queries. You still control the database engine, parameters, and data.

 **Mental model:** RDS is "**managed SQL**." AWS owns the host, OS, and DB engine maintenance; **you** own the data, schema, queries, and tuning. (You do **not** get OS/SSH access.)

Common uses: transactional apps (OLTP), web/mobile backends, e-commerce, SaaS, ERP, anything needing a managed SQL database with HA and backups.

2. Supported Engines

Engine	Notes
Amazon Aurora	AWS-built, MySQL- & PostgreSQL-compatible; cloud-native performance & scaling.
MySQL	Popular open-source.
PostgreSQL	Feature-rich open-source.
MariaDB	MySQL fork.
Oracle	Commercial; BYOL or license-included.
SQL Server	Microsoft; multiple editions.

Aurora is technically part of the RDS family but is a distinct, re-architected engine — see [section 9](#).

3. Core Concepts

Concept	Description
DB Instance	The running database environment (compute + memory + storage).
DB Instance Class	The hardware size (e.g. <code>db.t3.medium</code> , <code>db.r6g.large</code>).
DB Engine	The database software (MySQL, Postgres...).
Endpoint	DNS name + port apps connect to.
Parameter Group	Engine configuration settings.
Option Group	Extra engine features (e.g. Oracle TDE).
Subnet Group	The VPC subnets (across AZs) RDS can use.
Multi-AZ	Synchronous standby for failover (HA).
Read Replica	Async copy for read scaling.

4. High Availability (Multi-AZ)

- **Multi-AZ deployment** maintains a **synchronous standby** replica in a **different AZ**.
- On failure (AZ outage, instance crash, maintenance), RDS **automatically fails over** to the standby — the **endpoint stays the same** (DNS just repoints), typically in 60–120 s.

- The standby is **not** readable (it's for HA, not scaling) — *(except Multi-AZ DB cluster mode, which adds 2 readable standbys).*
- **Multi-AZ ≠ backup and ≠ read scaling.** It's purely about availability/durability.

5. 📖 Read Replicas (Scaling Reads)

- **Asynchronous** copies you can **read from** — offload reporting/read traffic from the primary.
- Up to **15** read replicas (engine-dependent); can be **cross-AZ and cross-region**.
- Each replica has its **own endpoint**; apps must be pointed at it deliberately.
- A replica can be **promoted** to a standalone primary (DR or migration).
- **Replica lag** is possible (async) — don't use for read-after-write consistency.

Multi-AZ vs Read Replica: Multi-AZ = synchronous, automatic failover, not readable (HA). Read Replica = asynchronous, manual use, readable (scaling). You often use **both**.

6. 💾 Storage Types

Type	Class	Use
General Purpose SSD (gp3/gp2)	SSD	Most workloads; gp3 lets you scale IOPS/throughput independently.
Provisioned IOPS (io1/io2)	SSD	I/O-intensive, latency-sensitive production DBs.
Magnetic	HDD (legacy)	Backwards compatibility only — avoid.

- **Storage autoscaling** can grow storage automatically as you fill it.
- Storage is **EBS-backed**; you can scale storage up (not down) without downtime.

7. 📀 Backups, Snapshots & PITR

- **Automated backups** — daily full + transaction logs; retention **0–35 days**. Enable **Point-in-Time Recovery (PITR)** to restore to any second in the window.
- **Manual snapshots** — user-triggered; kept until you delete them (survive instance deletion).
- Backups/snapshots are stored in **S3** (managed by AWS) and can be **copied cross-region** and **shared** across accounts.
- Restoring a snapshot **creates a new instance** (you then repoint the endpoint).

⚠️ Automated backups are deleted when the instance is deleted (unless you take a final snapshot). **Manual snapshots persist** — use them before risky changes.

8. Security

- **Network:** launch in a **VPC**, control access with **Security Groups**; keep DBs in **private subnets** (no public access).
- **Auth:** native DB users/passwords, or **IAM database authentication** (token-based), or **Kerberos/AD**.
- **Encryption at rest:** **KMS** (enable at creation — can't easily add later; encrypts storage, backups, snapshots, replicas).
- **Encryption in transit:** SSL/TLS to the endpoint.
- **Secrets:** store/rotate credentials in **Secrets Manager** (native RDS rotation).
- **Audit:** CloudTrail (API) + engine audit logs to CloudWatch.

9. Amazon Aurora

AWS's cloud-native engine, MySQL- and PostgreSQL-compatible:

- **Storage auto-grows** up to 128 TiB; data is **6-way replicated across 3 AZs** automatically.
- Up to **15 low-latency read replicas**; failover in seconds via a **cluster (writer/reader) endpoint**.
- **Aurora Serverless v2** — autoscales capacity (ACUs) up/down with load; great for variable/spiky workloads.
- **Global Database** — sub-second cross-region replication for global apps & DR.
- Typically **3–5x MySQL / 2–3x PostgreSQL** throughput; pay for storage actually used.

Choose **Aurora** for performance, automatic storage scaling, and fast failover; choose **standard RDS engines** for exact version parity, lower entry cost, or specific feature needs.

10. Maintenance & Scaling

- **Maintenance window** — weekly window for patching; some changes need a brief restart.
- **Vertical scaling** — change the **instance class** (brief downtime, or use Multi-AZ to minimize).
- **Storage scaling** — increase storage / IOPS online; **autoscaling** optional.
- **Read scaling** — add **read replicas**; **Aurora Serverless** scales compute automatically.
- **Blue/Green Deployments** — create a synchronized staging copy, test, then switch over with minimal downtime.

11. Monitoring

- **CloudWatch metrics** — CPU, connections, freeable memory, IOPS, storage, replica lag.
- **Enhanced Monitoring** — OS-level metrics at up to 1-second granularity.
- **Performance Insights** — visual DB load + top SQL/wait analysis for tuning.
- **Event subscriptions** (SNS) — failovers, low storage, backups, etc.
- **CloudTrail** — management API audit.

12. 💰 Pricing

You pay for:

1. **DB instance hours** — by class & engine; On-Demand or **Reserved Instances** (1–3 yr, big savings).
2. **Storage** — per GB-month provisioned (+ provisioned IOPS for io1/io2).
3. **Backup storage** — free up to DB size; extra beyond that.
4. **Data transfer OUT** and **cross-region** replication.
5. **Multi-AZ** roughly doubles instance/storage cost (standby).
6. **Extras** — Performance Insights (long retention), snapshots export, etc.

🔑 **Cost tips:** use **Reserved Instances** for steady prod · right-size the class · **Graviton (g)** instances for price/perf · stop non-prod (up to 7 days at a time) · trim old snapshots · use **Aurora Serverless** for spiky/dev workloads.

13. 📋 Common CLI Commands

```
aws rds create-db-instance --db-instance-identifier mydb \
  --db-instance-class db.t3.micro --engine postgres \
  --master-username admin --master-user-password '****' \
  --allocated-storage 20 --vpc-security-group-ids sg-123 --multi-az

aws rds describe-db-instances
aws rds create-db-snapshot --db-instance-identifier mydb --db-snapshot-identifier snap1
aws rds restore-db-instance-from-db-snapshot \
  --db-instance-identifier mydb-restored --db-snapshot-identifier snap1
aws rds create-db-instance-read-replica \
  --db-instance-identifier mydb-replica --source-db-instance-identifier mydb
aws rds modify-db-instance --db-instance-identifier mydb \
  --db-instance-class db.r6g.large --apply-immediately
aws rds reboot-db-instance --db-instance-identifier mydb --force-failover
aws rds delete-db-instance --db-instance-identifier mydb --final-db-snapshot-identifier final-snap
```

14. RDS vs Other AWS Databases

Service	Type	When
RDS	Managed relational (SQL)	Traditional OLTP apps needing a specific engine.
Aurora	Cloud-native relational	High performance/scale relational with auto-storage & fast failover.
DynamoDB	Managed NoSQL (key-value/document)	Massive scale, single-digit-ms, serverless, flexible schema.
ElastiCache	In-memory (Redis/Memcached)	Caching, sessions, leaderboards, sub-ms reads.
Redshift	Data warehouse (OLAP)	Analytics over huge datasets.
DocumentDB / Neptune / Keyspaces / Timestream	Purpose-built	Documents / graphs / Cassandra / time-series.

15. Quick Mental Model

- **RDS = managed SQL.** AWS runs the host/OS/engine maintenance; you own data + queries (no SSH).
- Pick an **engine** (MySQL, Postgres, MariaDB, Oracle, SQL Server, or **Aurora**).
- **Multi-AZ** = synchronous standby for **automatic failover** (HA), same endpoint — *not* readable, *not* a backup.
- **Read Replicas** = async, readable copies for **scaling reads** (own endpoint, possible lag).
- **Backups** (automated + PITR) and **manual snapshots** (persist) protect data — snapshot before risky changes.
- Keep it **private in a VPC**, encrypt with **KMS**, manage creds in **Secrets Manager**.
- **Aurora** for cloud-native scale/perf; **Reserved Instances + right-sizing** for cost.

 Notes compiled as a quick reference for Amazon RDS.